

I. INTRODUCTION

This is for...

II. EXTENDED LITERATURE REVIEW

A. Complexity of Clinical Narratives

Clinical letters use narrative language so that clinicians can record the patient's situation, their judgement, and their uncertainty. That flexibility supports clinical communication, but it makes the information difficult to use at scale [1]. Clinicians often use hedging to describe certainty, frequency, and quantity rather than stating a precise value [2]. The task of information extraction (IE) is to convert that nuanced information into a standardised form while preserving its essential meaning.

Epilepsy clinic letters illustrate that complexity. One letter can contain diagnoses, prescriptions, history, seizure descriptions, and several references to seizure frequency. Fonferko-Shadrach *et al.* report much lower human agreement for seizure frequency (per-item F1 = 0.47) than for prescriptions (0.87) or diagnoses (0.83) [3]. They attribute this to the variety of frequency expressions and to mentions of more than one seizure or event type. Effective extraction requires clearly defined policies and procedures describing what to extract and how to represent it.

B. Seizure Frequency as a Selection Task

The same letter can be used for different purposes. ExECT and the later annotation study treat it as an inventory: they record several entity mentions and their attributes. One letter may list several anti-seizure medications, each with a name, dose, unit, and administration frequency [3], [4]. That form is appropriate when the aim is building a broad clinical inventory.

The task in this study has a different purpose. Gan *et al.* assign each letter one label for the patient's current seizure frequency status [5]. The system has to gather candidate descriptions, interpret their meaning and time frame, and choose the single normalised label that best represents the current pattern. Earlier one-label studies use the same formulation [6]–[8]. The single label extraction implies that it is all one task, but this paper proposes two separate stages: extracting every candidate with its evidence in a canonical form, then deciding the final answer from those candidates.

Two common cases show why that split matters. Diary-style counts may need to be combined: *1 in June, 0 in July, 2 in August* can become *3 in 3 months*. Competing statements may instead require precedence: *She may remain seizure-free for up to 4 months, but then will experience clusters of 5 seizures in a single day*. Different annotation guidelines would record those cases differently. Separate stages make the policy choice inspectable rather than implicit.

C. Rules, Fine-Tuned Models, and LLMs

Several papers have published seizure frequency extraction systems using rules, fine-tuned models, or prompted large language models. Rules match terms and apply domain logic. ExECT does this for a clinical inventory; Decker *et al.* use regular expressions for seizure-related variables [4], [9].

Fine-tuned BERT models learn the mapping from labelled examples, as in Xie *et al.* and Fernandes *et al.* [6], [10]. Table VI lists the F1 scores those studies report. The scores are not a ranking of methods: each study uses its own dataset, label set, and evaluation.

Prompted and fine-tuned LLMs are more recent. Holgate *et al.* show that a frozen model can extract seizure information from instructions and examples [7]. Later work, including Holgate *et al.* and Gan *et al.*, fine-tunes a model to a defined output format [5], [8].

BERT-style encoders and decoder-based LLMs share the transformer architecture. The practical difference is in-context learning: a LLM can follow instructions and examples without a task-specific training run [11]. That behaviour transfers to clinical extraction, where clear instructions and examples can reduce the need for heavy post-processing [12]. Gan *et al.* show that combining structured prompting and fine-tuning can deliver even stronger performance. This paper explores whether structured prompting in a modular system can deliver comparable performance to a fine-tuned LLM.

D. Modular and Sequential Extraction Pipelines

Multi-stage clinical extraction pipelines split a task into distinct operations instead of asking a model for one final answer in a single step. CLINES uses stages for chunking, extraction, attribute assignment, normalisation, date resolution, and aggregation [14]. Seizure frequency studies have also decomposed the task in structured prompting including numbered reasoning steps and summarise-then-extract designs [7], [15].

This study decomposes the task into two stages: extracting candidate evidence into a structured record, and deciding the final answer from that record. One LLM call performs the extraction. The decision stage is performed in two ways on the same saved record, by recorded rules and by a second LLM call, so the two executors can be compared with the extraction held fixed. Past research on seizure frequency gives rules only a repair role after an LLM; this paper gives them the decision role.

III. PLANNING

The project lifecycle is divided into five stages: planning, development, implementation, testing and evaluation. The sections below describe the key decisions, the rationale behind them, and the sources they were based on, in chronological order as far as possible.

There are two key documents in the planning phase:

- 1) The project brief, created by my supervisor.
- 2) The problem specification and project plan, created by me.

A. Project Brief

The project brief outlines the initial scope of the project. The problem specification and project plan refines the scope, specifies the research objectives and describes the methods used to achieve them. The project brief was:

Instead of training models, this project builds a training-free (or minimal-training) multi-agent extraction system that reads epilepsy letters and outputs structured fields (e.g., ASM, seizure type/frequency, investigations, epilepsy syndrome/type). The student will design roles such as: (a) Section/Time-line Agent (segments text, builds timeline), (b) Field Extractor Agents (one per field group), (c) Verification Agent (checks evidence spans, contradictions, missingness), and (d) Aggregator Agent (produces final JSON + confidence + citations to text spans). The key research goal is reliability: compare single-prompt extraction vs multi-agent extraction under the same budget constraints, and test improvements from self-consistency, evidence requirements (“answer only if supported by quote”), and structured output validation. Evaluation uses field-level accuracy/F1, and robustness tests. Data can be synthetic or de-identified samples if available. Deliverables: an end-to-end agentic pipeline, evaluation harness, and a dissertation focused on engineering + empirical study of reliability.

The deliverables are a narrow specification of what a successful project would look like:

- 1) An end-to-end agentic pipeline.
- 2) An evaluation harness.
- 3) A dissertation focused on engineering + empirical study of reliability.

The type of data this would be applied to was narrowly defined as epilepsy letters, but it was undetermined whether the specific data source was synthetic or de-identified samples. This was one key decision to be agreed in the project kick-off meeting.

The research would compare a single-prompt extraction vs. multi-agent extraction, and the multi-agent system included a number of initial suggestions such as verification agent and aggregator agent. The precise implementation of the multi agent system would be defined as the research progresses, guided by best practices in the literature and the results from empirical experiments.

In the initial project kick-off meeting it was agreed that the specific data source used would be a set of 1,500 synthetic clinical letters that the supervisor had access to. As these were synthetic letters, there would be no need to get specific ethics approval to use for research purposes.

B. Problem Specification and Project Plan

The problem specification and project plan was uploaded to the university Canvas site on the 5th June 2026. This is a direct extract from the methods section:

There are two primary data sources in this project:

- 1,500 synthetic clinical letters and structured labels from Gan *et. al* [5]
- 200 synthetic clinical letters and structured labels from Fonferko-Shadrach *et. al* [3]

The project will use LLMs as the core AI tool, using both open (e.g. Qwen 3.6) and closed (e.g. GPT 5.4) models. The research will involve building an agent harness around the model that enables it to ingest clinical letters and prompt instructions, produce a structured output, and have that output validated and evaluated.

The research will be empirical in nature and focus primarily on assessing the quality, reliability and validity of the outputs. Cost, latency, data schema complexity and evidence specificity will be key experimental variables tested.

Evaluation will take place in three key forms: comparing the accuracy (F1 score) of the outputs vs. the ground-truth labels from the two data sources, assessing the validity of the structured data schema created, and assessing the validity of the evidence cited. A series of ablations will be conducted to isolate the impact of different components of the agent harness.

The F1 score evaluated on a set of ground-truth labels is the de-facto standard for evaluating AI models on how well they can complete a classification task. Evaluating the model on two different datasets optimised for two different tasks effectively tests the flexibility of the system.

The ExECTv2 dataset [3] was used extensively as per the original plan. The pipeline was built, experiments were conducted and results were analysed in depth and benchmarked against the prior results. However, when the time came to write the paper it became clear that writing about two datasets with different annotation policies, evaluation measures, error profiles and sample sizes required the reader to hold too much information in mind.

As a result, the empirical analysis in the paper and in these materials focus exclusively on the seizure frequency extraction task. However, we used the pipeline built for ExECTv2 to extract diagnoses, medications and investigations from the main dataset of 1,500 synthetic letters as a feasibility study. Examples of that are shown in the appendices here, and some of the design choices for the seizure frequency extraction pipeline are directly informed by designing a modular system that can be applied to both tasks. Results benchmarked against the ExECTv2 dataset will be published in a separate paper.

The following sections describe how the rest of this plan was implemented.

IV. DEVELOPMENT

This section explains how the initial project proposal was translated into a focused and testable research approach. It records the evidence, design choices, and trade-offs that led to the final method.

A. Original Design Requirements

The original brief contained a series of key design requirements for the pipeline. It must:

- 1) Ingest epilepsy letters
- 2) Output validated structured fields
- 3) Contain verified evidence citations
- 4) Produce this in an aggregated JSON

These design requirements were tightly coupled to a multi-agent framework, e.g using a *Verification Agent* to check evidence spans and an *Aggregator Agent* to produce the final JSON and citations.

The literature review showed that multi-agent systems are particularly useful for complex tasks [25], especially on long-context tasks [26]. However, research also shows that single-agent systems are often as effective when they are given equal compute and multi-agent systems create new failure modes as errors propagate throughout the system [26].

For this task, the full epilepsy letter and instructions fit comfortably in the LLM’s context window so there is no need for convoluted multi-hop pipelines or document splitting that multi-agent systems performed particularly well on [14]. Thus, a multi-agent system can not be justified on that basis.

It is less clear whether this task is sufficiently complex to justify a multi-agent system. Past research has shown that LLMs struggle with temporal logic and numerical logic, and both occur in this task. It is also a task that human annotators struggle to do reliably and consistently [3]. This suggests the task is at least somewhat complex and a multi-agent system is justified on that basis.

However, it is important that each LLM call is given a sufficiently complex task to justify the added system complexity. A key goal of the research is to design a reliable system. A reliable system is one that contains the minimum amount of complexity necessary to function. More components means creates more failure modes, and in a sequential process those errors propagate and multiply at each step.

B. Deterministic Code for Deterministic Tasks

A key design choice early on was to simplify. Any task that can be reliably completed by a deterministic system (code) instead of a probabilistic system (LLMs) should be done deterministically.

Evidence verification and schema validation are two tasks that code can do reliably and efficiently. Initial experiments showed that LLMs can not do this task reliably; they introduce as many new schema errors as they repair, and they occasionally miss invalid evidence. However even if they could do the task reliably, code already does this at a much lower cost and latency.

Therefore a dedicated *Aggregator Agent* to produce the final JSON with text citations and a dedicated *Verification Agent* to check evidence spans were removed from the design after initial experiments. A *Field Extractor* agent was no longer relevant as the task was limited to a single field, seizure frequency. Initial experiments showed that a *Section/Timeline Agent* was a net-negative contribution because correct seizure frequency references were not always in the dedicated seizure frequency section, and splitting up the text encouraged the LLM to select wrong answers more frequently.

C. Pipeline Design Review

After initial experiments, all four of the original proposed LLM agents had been ruled out because they were now either invalid or empirically performed worse than a single LLM. This led to a major review of the pipeline design.

Comprehensive error analysis showed that the LLM was consistently finding valid seizure frequency patterns, providing exact evidence from the text and often choosing defensible answers based on the instructions. It did not need another LLM call to check its evidence or structure it into JSON; it needed cleaner task separation.

The task of extracting the correct seizure frequency label is composed of three sub-tasks: finding relevant mentions of seizure frequency in the text, encoding that information into the form the gold-standard labels expect, and applying a policy to choose which of those labels matches the gold-standard selected label. The labelling conventions and policy choices in the gold-standard evidence are valid, but they are not the only valid choice. Therefore a key pipeline requirement is producing results that conform to an explicit set of subjective choices.

The original brief separated mechanical tasks across multiple LLM calls; one call splits the evidence, another checks the evidence, another aggregates it together. With simplicity and efficiency in mind, it would be preferable if a single LLM call could do all of those tasks at once. This would require the LLM to generate a much richer intermediate schema. Rather than extracting a single seizure frequency label, the LLM would extract every relevant seizure frequency pattern as a distinct entity in a nested JSON, with exact evidence for every entity, and make an initial proposal.

Using this richer schema, a second LLM call could then review the evidence and apply the correct policy in hard cases. This reframes the decomposition around clinical tasks (extract the candidates, then decide between them) instead of mechanical tasks. The richer record also makes it possible for deterministic rules to apply the same policy to the evidence the LLM extracted.

Past research shows that deterministic code can reliably normalise seizure frequency values, and hybrid LLM and rules systems can outperform LLMs alone. Rules-based systems struggled with seizure frequency extraction more than medications, but they still performed better than human experts. Deterministic code already performed schema validation and evidence verification more reliably than LLMs, so there was a case for more deterministic code in the pipeline.

D. Final Pipeline Design

The final outcome of this design review was a pipeline with two paper-facing stages, *extract* and *decide*. The implementation names three stages, *find*, *encode* and *select*: the cited extraction prompt already writes labels in the gold standard’s form, so *find* and *encode* are bundled into one call and reported as *extract*; *select* is reported as *decide*. Those implementation names are kept here only where they identify an existing artifact.

The paper’s core comparison holds the extraction call fixed and changes only who performs `decide`: recorded rules (*Hybrid*) or a second LLM call (*LLM-only*). Because the decision stage reads the saved record, both executors replay the same model output without a new call, and every difference between them is attributable to the decision stage.

During development, rules were also implemented for the extraction role, and an end-to-end rules-only baseline was built and scored. That baseline is not part of the dissertation comparison. It remains in the repository as development history with its own recorded protocol; it is not reinterpreted here. The other decompositions of the LLM work that were tried are summarised in the Evaluation section as secondary experiments with their evidence retained.

The original project brief required a pipeline that could ingest epilepsy letters, output validated structured fields, contain verified evidence citations and produce this in an aggregated JSON. It required an evaluation harness and a dissertation focused on engineering and an empirical study of reliability. Each requirement maps onto a mechanism and its evidence: validated structured fields to the schema-validated record (Table I); verified evidence citations to the exact-quote obligation and the exact-evidence rate; aggregated JSON to the final record; and the empirical study of reliability to the two-executor comparison, the prompt ablations, the six-model comparison and the sampling ablations reported in the paper.

The two-executor comparison keeps the empirical core on the brief’s question. It asks whether a second LLM call is needed to apply the policy reliably, or whether recorded code does the same work from the same record, and it answers with a paired test on held-out letters.

V. IMPLEMENTATION

The implementation was designed to produce a structured, reproducible account of current seizure frequency from a clinical letter, while retaining the evidence and intermediate choices needed to inspect how an answer was reached. It describes the reusable pipeline and the controlled comparison design; performance results and the findings of individual ablations are reported in the evaluation section.

A. Structured Record

Each configuration receives a synthetic clinical letter and produces a structured seizure frequency record. The final record contains the submitted frequency label, direct supporting evidence, and the metadata needed to identify the method, model configuration, data split, and scoring policy used. This design makes the final answer usable for benchmark evaluation without reducing the whole process to an opaque single-label extraction.

The pipeline uses a richer intermediate record than the final benchmark label. During extraction, it can retain multiple candidate seizure frequency events, their source spans, raw wording, time information, and an initial proposed answer. This is important because a letter may contain several plausible statements: historical and current rates, periods without

seizures, isolated breakthrough events, clusters, and uncertain or qualitative descriptions. Preserving the alternatives avoids losing information before the annotation policy has been applied.

B. Extract and Decide

The record distinguishes the source wording from its encoded representation and from the final selected answer. The decision stage can revise the provisional answer while retaining the evidence on which the original proposal was based. The final output is validated against the expected schema, while source data and raw model output are retained separately from format repair and clinical policy decisions.

The interface between the two stages is fixed. `Extract` receives the full letter and returns the candidate record: every seizure frequency event with its quoted span, category and canonical label, plus a provisional answer. `Decide` receives that record and the decision policy, never the letter, and returns the final label with the event(s) that support it. In the implementation, `extract` is the `find` stage whose prompt already writes the gold standard’s label forms (so the `encode` step is bundled into the same call), and `decide` is the `select` stage. A small deterministic `encode` step still runs between them in *Hybrid* to rewrite any label the model left in a non-canonical form; on the held-out set it changes at most 0.01 (355 to 360 correct letters before rules `decide`).

Each stage produces an inspectable output. This is particularly useful when the final answer differs from the provisional one: the change can be attributed to the evidence extracted or to the policy decision. It also permits the decision stage to work from the same saved candidate record, rather than requiring every alternative implementation to make a new model call. This significantly sped up the optimisation process and is what makes the two-executor comparison a paired one.

C. Rules and Deterministic Components

Deterministic components implement tasks for which explicit, reproducible handling is appropriate. These include parsing and schema repair, validation of required evidence fields, aggregation of the final record, and rules for normalising frequency expressions into the task’s closed vocabulary. The implementation keeps format-only repair, benchmark representation, and clinically meaningful policy rules distinct so that a change to one does not silently alter another.

In *Hybrid*, recorded rules perform the decision role. Their authority is recorded by rule type: whether a rule merely standardises a representation (dialect or `encode`), blocks an unsupported provisional answer (`gate`), rewrites the submitted label from the extracted evidence (`rewrite`, e.g. totalling a seizure diary), or reselects a different already-extracted event (`reselect`). No decision rule scans the letter for a new candidate. This makes deterministic processing an explicit part of the method rather than unreported post-processing. Rules for the extraction role were also implemented during development; they are repository history, not part of the reported comparison.

D. LLM Components

The LLM implementation uses structured prompts that ask the model to return a record containing candidate seizure-frequency events, evidence, and a proposed answer. The extraction call reads the letter and returns that record. In `LLM-only`, a second call receives the record and adjudicates between competing candidates under the same policy that the rules apply, written as instructions with worked examples. Neither call uses tools, chooses the next step or invokes another model; each is a prompt with a fixed input and a schema-validated JSON output, which is why the system is described as a fixed pipeline of LLM calls rather than as agents.

Holding the extraction call fixed and changing only the decision executor provides a controlled way to compare rules and LLMs on the substantive clinical decision, while deterministic validation and output assembly remain common infrastructure.

VI. TESTING

Testing verified that the implemented system and research controls behaved as intended before its comparative results were interpreted.

Data-loading tests verify that source records, gold labels, evidence references, quality flags, and label normalisation are preserved. The test suite also checks that the fixed split manifest contains the intended disjoint training, development, and held-out partitions, and that loaders return records in the manifest-defined order. These checks protect against accidental changes to the dataset surface, label interpretation, or the membership of a reported split.

The split policy was treated as a testable research safeguard rather than a documentation convention. Development work used the permitted development partition for prompt and rule changes, while the held-out partition could be used only under a fixed protocol and reported at aggregate level. Tests enforce that held-out replay artifacts do not retain row identifiers, source notes, labels, predictions, or error lists.

Focused regression tests capture failure modes identified during development. These include aggregating diary entries across months, avoiding double-counting when the same evidence is repeated or split across candidates, handling zero-count months, interpreting clusters, and distinguishing current seizure burden from historical or less relevant statements. Each fixture represents a named clinical or annotation-policy obligation, rather than a generic collection of examples. These tests were used whenever a prompt, rule, parser, or stage boundary changed. Their purpose was to show that a local improvement did not silently regress previously correct behaviour.

The implemented methods were tested as pipelines as well as individual functions. Prompt-contract tests check that model requests preserve the required structured fields and do not inadvertently introduce irrelevant metadata. Pipeline and runner tests check that the intended components run in the intended order and that their outputs are written into the common record structure.

Together, these tests verify that the implementation produces valid records, applies the intended data and split controls, and preserves known clinical-policy behaviour.

VII. EVALUATION

This section evaluates whether the verified pipeline met the research objective: to extract a patient’s current seizure frequency burden from a clinical letter reliably, with a structured evidence record and a reproducible comparison between rules and LLMs.

All held-out (`test450`) figures in this section are aggregate totals; no held-out letter or output was inspected. Development (`dev750`) figures may be accompanied by row-level mechanism examples. Unless stated otherwise the model is Gemini 3.7 Flash at temperature 0 with low thinking effort, and the metric is Purist micro-F1 with correct letters in parentheses.

A. Extracting Multiple Candidates with Exact Evidence

A key feature of the pipeline is the rich extraction record. The model is asked to extract every seizure frequency pattern in the text, quote an exact evidence span for each, write each in a canonical form, and propose which best represents the patient’s current status. That record is what allows the second LLM call in `LLM-only` and the rules in `Hybrid` to re-use the evidence and improve on the provisional answer by +0.06 to +0.07. The cost is a more complex schema, and this is where the local models fall slightly behind.

The gold standard records one label and one quote per letter. The extraction record holds more: on `test450`, Gemini extracted 985 events over 450 letters (2.19 per letter) and the six models ranged from 2.19 to 2.82 events per letter. On `dev750` the Gemini mean is 2.16 events (median 2, range 1–6), while the set of events supporting the final answer has a median of 1. The record therefore preserves alternatives that a one-label output would discard.

Table I gives each model’s adherence to the extraction contract on `test450`. Every model returned a parseable record for at least 442 of 450 letters and no call failed. The four hosted models quoted an exact substring of the letter for the selected event in 99% or more of letters; Qwen (97%) and Gemma (93%) occasionally paraphrased. Gemma also needed the most non-semantic format repair (15 letters), typically doubled braces or quotation marks that broke the nested JSON, and left 8 letters unparsed after repair; those letters are scored as incorrect. Format repair is recorded separately from every semantic rule and never changes a label.

B. Extraction Prompt Variants

The paper’s Table VII compares the full extraction prompt with exactly three focused variants, each removing one ingredient while the others remain: the worked examples, the closed list of allowed label forms, and the evidence obligation (the evidence fields and the exact-quote instruction, removed together as one package). Table II repeats those rows with the intermediate encode stop exposed, so the reader can see that the deterministic label rewrite between `extract`

TABLE I: Six models as the extraction call on `test450`: scores and contract adherence (Hybrid)

Model	Purist	Prag.	Exact ev.	Repairs	Unparsed	Retry rej.	Events
Gemini 3.7 Flash	0.860 (387)	0.880 (396)	99.8% (449)	0	1	0	985
Grok 4.6	0.853 (384)	0.889 (400)	100% (450)	0	0	0	1063
DeepSeek V4 Flash	0.820 (369)	0.849 (382)	99.6% (448)	2	0	0	1267
GPT-5.6 Luna	0.789 (355)	0.820 (369)	98.9% (445)	0	1	0	1102
Qwen 3.8 27B	0.762 (343)	0.807 (363)	96.7% (435)	3	2	0	1004
Gemma 4 26B	0.724 (326)	0.773 (348)	92.9% (418)	15	8	7	1068

Exact ev.: the quoted span for the selected event is an exact substring of the letter. Repairs: letters needing a non-semantic JSON dialect repair. Unparsed: no valid record after repair. Retry rej.: format-only retries whose output was rejected. Events: extracted events over 450 letters. Source: `paper_experiments/gan/runs/{model}/test450/comparison.json`.

TABLE II: Extraction prompt ablations on `test450` (Hybrid replay)

Extraction prompt	Provisional	After encode	Final
Full codebook prompt	0.789 (355)	0.800 (360)	0.860 (387)
Without examples	0.767 (345)	0.776 (349)	0.822 (370)
Without closed allowed-label forms	0.771 (347)	0.776 (349)	0.809 (364)
Without evidence obligation	0.767 (345)	0.771 (347)	0.822 (370)

and `decide` moves at most five letters in any row. The encode column is the shared replay, not a fourth ablation. The decrements are not additive.

Two further prompt families were run and are kept as secondary evidence because they change more than one thing at once or use a different scorer, so they are not comparable with Table II.

Source-near wording. A variant of the extraction prompt (`gan_llm_extract_raw`) keeps the same event schema and policy but drops the closed label list and asks for letter wording or a near-source phrase. Its provisional answer is 0.547 (246); a selected-evidence encoder that rewrites the chosen span into the gold form raises this to 0.744 (335), and rule `decide` to 0.793 (357). It exists to test whether the model should write the gold form itself or leave that to code. The bundled codebook prompt is better at every stop, so the cited prompt asks for the canonical form directly.

Holgate-style ask. Two variants reproduce the three-step instruction of Holgate *et al.* [7] and are scored with a dialect map (`holgate_dialect_v1`) that accepts their label conventions. Wrapped in this paper’s event-and-selection schema with the quote obligation, the ask scores 0.616 (277) provisional, 0.640 (288) after encode and 0.649 (292) after `decide`. Asked for one label with no schema or quote requirement, it scores 0.440 (198) and produces no record to `decide` from. Comparing the two Holgate rows isolates the structured record plus evidence obligation with the clinical ask held fixed; they do not isolate the instruction rewrite alone, and their scorer differs from the main table.

C. Secondary Decompositions of the LLM Work

The brief asked whether more LLM calls make extraction more reliable. The following decompositions were built and measured. None replaces the two-stage design and they are

reported here, with their scope, so that the negative and null results are on record.

A separate label-rewriting call. Inserting a second LLM call that rewrites each extracted event into the gold form, before a third call decides, made the score worse. On `test450` the codebook extraction stop of 0.79 (354) fell to 0.65 (291) after the rewriting call and recovered only to 0.71 (320) after the deciding call, with the development split showing the same shape (0.78, 0.67, 0.76). The development mechanism study found 89 harmed and 21 rescued letters at the rewriting call. A later replay with the living decision prompt confirmed the drop (0.68, 304) and showed the living `decide` call recovering to 0.86 (386), one letter below `Hybrid`. The rewriting call does not re-read the letter and overwrote answers that were already correct. This is why `LLM-only` is two calls, not three.

One call for everything. After the main comparison was fixed, a single prompt that states the decision policy and its worked examples inside the extraction call was run as an ablation. Scored at its own provisional answer, it reaches 0.87 (392) on `test450` and 0.88 (657) on `dev750`, one letter above `Hybrid` on development and five above on holdout, with no parse or call failures. It is reported as an ablation, not as a headline row: it was not part of the predeclared comparison and has not been promoted. It shows that the policy can also be applied inside the extraction call. The case for the separate decision stage is not a score margin; it is that the decision is inspectable, replayable without a model call, and changeable without re-prompting.

Verification then correction. A development-only variant asked a second call to verify the provisional label against its quoted evidence and correct it. It removed true positives, because the gold label is a labelling convention rather than a paraphrase of the quote (*multiple per week* against *many events in the last week*). A related generate-then-verify program optimised with DSPy did not beat the development ceiling of the time (0.731). No clean held-out artifact exists for this variant.

Section or timeline splitting. A development-only variant split the letter into sections before extraction. It was a net negative: the correct seizure frequency reference was not always in the seizure-frequency section, and splitting encouraged the model to select wrong answers more often. No held-out artifact exists.

Sampling and orchestration on an older stack. A pre-

declared July study on a 50-letter development panel with gpt-4.1-mini compared greedy decoding with self-consistency (temperature 0.7, majority vote), a tool-using single call, and two multi-call orchestrations. Majority voting scored 15/50 against greedy 19/50. The multi-call variants scored 29–32/50 but with 4–8 losses against greedy and failed the predeclared gate (wins ≥ 5 and losses ≤ 1). The study used an older model, 50 rows and a gate designed for a different sample size. The permitted reading is narrow: on this task the greedy answer is already the modal answer, so sampling adds noise; and dynamic orchestration was not tested at scale. It does not support the sentence “multi-call pipelines do not help”.

D. Hybrid Pipeline Variants

The Hybrid row in the paper is one extraction call followed by a deterministic label rewrite and the decision rules. Two variants of that stack were measured on *test*₄₅₀. Skipping the label rewrite so that the decision rules act directly on the model’s own label forms scores 0.85 (382), five letters below the cited row. Adding rule-extracted candidates to the model’s record before the rules decide scores 0.82 (368), below the model’s record alone, so the cited row uses the model’s record without rule-extracted additions.

The decision rules are nine named families applied in a fixed order. On *dev*₇₅₀ they change the answer on 67 of 750 letters: 45 rescues, 4 harms, net +41 (608 to 649 before the final family was added). The largest families are diary totals (+25), dated sequences (+10) and post-treatment bursts (+7). The last family added, a last-event rule that rewrites a short numeric seizure-free span after a day-dated last event, was accepted on development evidence and replayed once on the held-out set as part of the final configuration (656/750 development, 387/450 held-out). Each family, its authority and its portability scope are listed in the repository rule catalogue.

VIII. DEVELOPMENT VS. TEST RESULTS

Figure 1 shows the two decision executors on both splits. Hybrid scores 0.87 (656/750) on development and 0.86 (387/450) on the held-out set, a generalisation gap of -0.01 . LLM-only scores 0.85 on both (640/750 and 383/450). Neither executor was tuned on held-out letters; prompts and rules were fixed on the development split and replayed once. The small gaps are consistent with the extraction call doing the pattern matching across the whole letter and the decision stage acting only on the refined record. The development-only rules-extraction baseline mentioned in the Development section showed the larger development-to-test drop typical of rules tuned on one corpus; it is not part of this comparison and is recorded in the repository.

In the early phases of development, comprehensive error analysis showed that the LLM was consistently choosing valid answers based on the instructions. For example, the LLM is told to choose the seizure frequency with the highest current or recent seizure pattern, if multiple seizure events are available. When it sees a pattern such as *2 in Jan, 0 in Feb, 0 in March, 5 in April* the model chooses 5 because

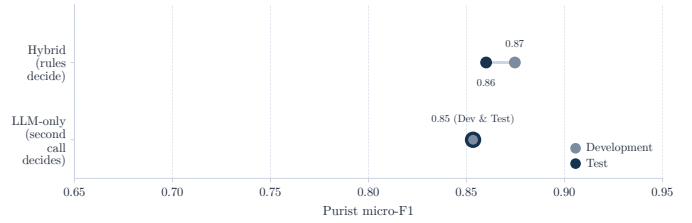


Fig. 1: The two decision executors on the development and held-out splits, Purist micro-F1. Both share the same extraction call. Hybrid moves from 0.87 to 0.86; LLM-only is 0.85 on both.

it is the largest and most recent pattern. However the gold-standard label is 7 in 4 months. When the prompt instruction was changed to add up all seizure events in a period, then it combined tonic-clonic seizures and generalised seizures over overlapping periods, while the gold-standard choose the highest count among the two seizure types. Separating these complex, somewhat contradictory instructions into a separate decision stage, whether a second LLM call or rules, produced the +0.06 to +0.07 improvement reported in the paper.

IX. EVIDENCE QUALITY BEYOND EXACT SPANS

The paper measures evidence quality only as exact-substring adherence (Table I). An exact span is not proof that the span semantically supports the label. The gold standard’s single reference quote is a loose comparator for that question: it may be exact, paraphrased, abbreviated, ellipsized or missing, so it is not ground truth for span correctness.

A development-first directional study is specified and drafted but not yet run. It measures two properties of the existing annotation reference on *dev*₇₅₀: whether it can be located as an exact source substring, and whether it conveys enough meaning to support the target label. It measures two matching properties of the pipeline record: whether the collected spans occur in the letter, and whether all raw grouped spans, considered together, support the target label under the decision policy. The adjudicator for the pipeline arm receives only the target label, the raw grouped spans and the policy; it does not see the letter, the normalised labels, the predicted answer or the annotation reference. It returns *supported*, *unsupported* or *insufficient*, a short reason, and controlled policy tags. A development pilot and an explicit acceptance decision must precede any aggregate-only replay on the held-out set. The study is a directional measurement and protocol check; it is not clinical validation.

The method is available as an open source package here along with a web app demonstrating key features of the pipeline here .

X. THE EXTRACTION PROMPT

The cited extraction request is the codebook prompt (implementation name `gan_llm_extract`). The object at the end of this document is the frozen template sent to the model: task, instructions, allowed label forms, event schema and selection

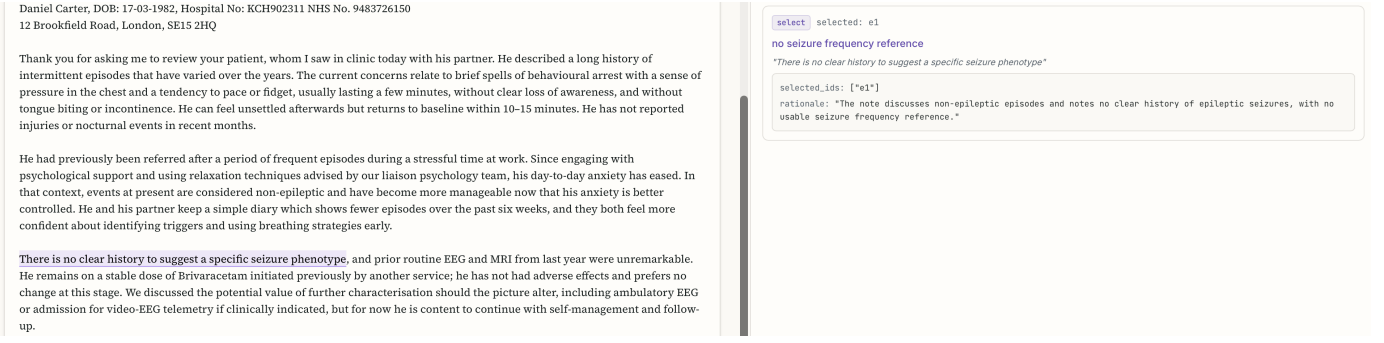


Fig. 2: Seizure free → unknown error example. Gold-standard label: seizure free for multiple month. Model selected no seizure frequency reference (unknown).

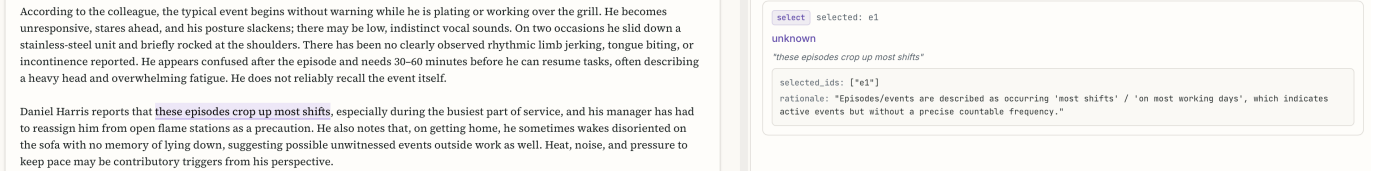


Fig. 3: Frequent → unknown error example. Gold-standard label: multiple per week. Model selected unknown.

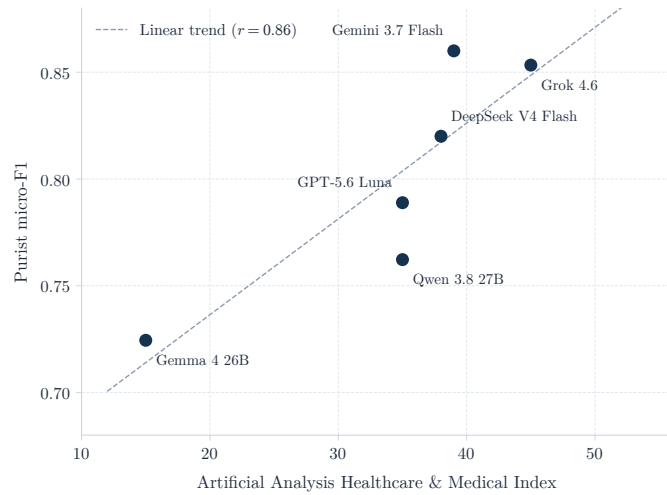


Fig. 4: The smartest models are the best frequency extractors.

TABLE III: Seizure frequency label forms

Family	Written forms
Rate	N per unit; N per N unit; N per N to N unit; N to N per unit; N to N per N unit; multiple per unit; multiple per N unit; N per multiple unit
Cluster	cluster per unit, N per cluster; cluster per unit, range per cluster; cluster per unit, multiple per cluster
Seizure free	seizure free for a duration
Unknown	unknown; no seizure frequency reference

schema. The four ingredients ablated in the paper map onto it as follows: the *instructions* are the task and instruction blocks; the *examples* are the worked examples inside the instructions; the *allowed labels* are the closed `label_forms` block; and

TABLE IV: Seizure frequency event categories in the extraction record

Category	Use
frequency_rate	A countable or range rate
cluster_frequency	How often clusters occur
seizure_free	A seizure-free statement
last_event_only	A dated last event without a rate
unknown_frequency	Seizures discussed, no usable rate
no_reference	No usable frequency evidence

TABLE V: Decision stage examples

Events	Provisional answer	After decide
3 in March; 6 in May	6 per month	9 per 3 month
Last seizure on 12th	seizure free	1 per month
March; well 3 weeks	for 3 week	

TABLE VI: Comparisons to past seizure frequency papers

Author	Year	Method	F1 score
Xie et al.	2022	Fine-tuned BERT	0.83*
Decker et al.	2022	Rules	0.82**
Fonferko-Shadrach et al.	2024	Rules	0.66
Fernandes et al.	2024	Fine-tuned BERT	0.37
Holgate et al.	2024	Prompted LLM	0.73
Holgate et al.	2025	Fine-tuned LLM	0.68
Gan et al.	2026	Fine-tuned LLM	0.81
Brown & Gan	2026	LLM and rules	0.86

the *evidence obligation* is the evidence fields of the event schema together with the exact-quote instruction. The letter is attached at run time as `note_text` and is omitted here. Research metadata such as `prompt_version` is not part of the model-facing payload.

Clinical Summary:

Since our last review, the patient describes meaningful improvement in day-to-day stability. Night-time childcare support has been introduced, sleep has become more consolidated, and both the patient and partner report steadier mornings with fewer interruptions. In this context, No further tonic-clonic seizures have occurred since 06/2017, although **three single jerks remain. These jerks were brief, without loss of awareness, and occurred during periods of sleep restriction earlier in the year.** No injuries, tongue biting, incontinence, or prolonged post-event confusion have been reported. The patient continues working and notes improved concentration since sleep regularised.

select selected: e2

unknown

"three single jerks remain. These jerks were brief, without loss of awareness, and occurred during periods of sleep restriction earlier in the year"

selected_ids: ["e2"]

rationale: "The patient is not completely seizure-free because three single jerks occurred earlier in the year, but the exact frequency is vague/not standardly countable."

Fig. 5: Infrequent → unknown error example. Gold-standard label: 3 per 14 month. Model selected unknown.

This template is not the source-near variant (gan_llm_extract_raw), which has no closed label_forms block, and it is not the second-call decision prompt used in LLM-only. Event categories are listed in Table IV and label forms in Table III.

XI. EXPERIMENTAL ENVIRONMENT

Experiments using local language models took place on a Dell XPS 16 9640 running Windows 11, with an Intel Core Ultra 9 185H, 32 GB RAM, and an NVIDIA GeForce RTX 4070 Laptop GPU (8 GB VRAM). Ollama v0.32.15 was used to serve the two local models (Qwen 3.8 27B, Gemma 4 26B).

Experiments using hosted APIs were run from a Mac mini M1 with 16 GB unified memory on macOS 26.5.2. API calls were made directly to the provider (OpenAI, DeepSeek) or through model routers (OpenRouter, AI Gateway). Four hosted models were used (Gemini 3.7 Flash, GPT-5.6 Luna, Grok 4.6, DeepSeek V4 Flash). Hosted accelerators are not disclosed by the providers.

All models used a temperature of 0 except Luna, which only accepts a temperature of 1. An ablation was run on Gemini using a temperature of 1. All models used a thinking or effort level of low. Ablations were run on Gemini using thinking levels of medium and high.

Python 3.14.4 was used to write the program. DSPy 3.2.1, Pydantic 2.13.4 and LiteLLM 1.87.1 were the core packages. DSPy co-ordinated the model calls and required structured JSON output. Pydantic validated that the structured output conformed to the schema. LiteLLM delivered API calls to Ollama and the model routers. Every reported cell is replayable from saved model output in the repository's paper_experiments/ directory without a new model call.

XII. DESCRIPTIVE CLINICAL-INVENTORY FEASIBILITY

The Gan gold standard is one current seizure-frequency label per letter. The same synthetic letters also mention diagnoses, medicines, investigations and several seizure frequency statements. To show the broader structured content available, a frozen four-family inventory schema and extraction program, built for the separate ExECTv2 task [3], was applied to a prespecified sample of 100 dev750 letters. The program was not tuned after the sample was drawn. No inventory gold exists on these letters, so Table VII reports only what the schema emitted. The counts are not precision, recall, accuracy or clinical validity; they are a descriptive feasibility panel that motivates later expert annotation on real correspondence. ExECTv2 benchmark results will be reported in a separate paper.

XIII. TOOLS DESCRIPTIONS

XIV. CRITICAL APPRAISAL

A. Successes

B. Challenges

REFERENCES

```
{
  "task": "Read the clinical note. Extract seizure-frequency facts as slim events, then select the current burden .",
  "instructions": [
    "Read the full clinical note and extract source-near seizure-frequency facts.",
    "Return events as slim clinical facts, not fully normalized answer records. Use raw_value for the text's stated rate, duration, last-event statement, or unknown/no-reference cue.",
    "Event kind must be one of frequency_rate, cluster_frequency, seizure_free, last_event_only, unknown_frequency, or no_reference.",
    "Use one no_reference event only when the note contains no usable seizure-frequency evidence. Do not use no_reference when seizures are discussed but frequency is unclear; use unknown_frequency instead.",
    "Keep seizure-free statements separate from unknown or last-event-only statements. Do not select seizure-free if other current seizure-like events remain active.",
    "Selection must choose the highest current or recent seizure burden across semiologies when several current seizure types are present.",
    "If the note gives an overall current seizure count plus a breakdown by seizure type, select the overall count for the seizure-frequency label rather than only the clinically most severe subtype count.",
    "Write the seizure-frequency label using only the allowed forms. Copy an example and change the numbers if needed.",
    "If the selected event has a countable stated rate, put the note's wording in raw_value and write the seizure-frequency label from the allowed forms.",
    "When the note says a last event occurred on a date and the patient has been well, stable, or seizure-free since, still extract the dated last-event fact as its own event even if the selection is seizure-free.",
    "When the note says a count such as 3 or 4 jerks occurred since a dated last tonic-clonic seizure, keep the source count and the dated anchor available in the event list.",
    "Every evidence value must be an exact substring from the note when possible.",
    "Return exactly one JSON object with no markdown."
  ],
  "label_forms": {
    "rules": [
      "Write the label using only the forms below. Copy an example and change the numbers if needed.",
      "Use digits, not word numbers.",
      "Use day, week, month, or year. Prefer the singular word.",
      "A night count is a day count: once per night becomes 1 per day.",
      "Flatten a bound: at most four per day, or ≤ four per day, becomes 4 per day.",
      "Do not write <=, >=, up to, at most, several, handful, frequent, or a calendar date.",
      "If none of these forms fit, write unknown when seizures are discussed, or no seizure frequency reference when there is no usable frequency evidence."
    ]
  }
}
```

TABLE VII: Descriptive inventory output on 100 dev750 synthetic letters

Family	Letters ≥ 1	Facts	Median (range)	Common subtypes
Diagnosis	76	130	1 (0–4)	Epilepsy
Prescription	81	204	2 (0–9)	levetiracetam, lamotrigine, sodium valproate, clobazam, topiramate
Investigations	40	68	0 (0–3)	MRI normal, EEG abnormal, EEG normal, MRI abnormal
Seizure frequency	50	81	0.5 (0–4)	seizures, seizure, increased, infrequent, frequent
Any family	97	483	5 (0–11)	—

```

},
"forms": [
  {
    "form": "N per unit",
    "description": "A single count in one day, week,
month, or year.",
    "examples": [
      "1 per day",
      "4 per week",
      "3 per month",
      "2 per year"
    ]
  },
  {
    "form": "N per N unit",
    "description": "A single count over a stated number
of days, weeks, months, or years.",
    "examples": [
      "1 per 2 day",
      "1 per 2 week",
      "2 per 6 month"
    ]
  },
  {
    "form": "N per N to N unit",
    "description": "A single count over a range of days,
weeks, months, or years.",
    "examples": [
      "1 per 2 to 3 day",
      "1 per 2 to 3 week",
      "1 per 4 to 6 month"
    ]
  },
  {
    "form": "N to N per unit",
    "description": "A count range in one day, week,
month, or year.",
    "examples": [
      "2 to 3 per day",
      "2 to 4 per week",
      "3 to 5 per month",
      "2 to 4 per year"
    ]
  },
  {
    "form": "N to N per N unit",
    "description": "A count range over a stated number
of days, weeks, months, or years.",
    "examples": [
      "2 to 4 per 3 month",
      "2 to 3 per 2 week"
    ]
  },
  {
    "form": "multiple per unit",
    "description": "More than one seizure in one day,
week, or month, with no number given.",
    "examples": [
      "multiple per day",
      "multiple per week",
      "multiple per month"
    ]
  },
  {
    "form": "multiple per N unit",
    "description": "More than one seizure over a stated
number of months, with no number given.",
    "examples": [
      "multiple per 2 month"
    ]
  }
],
"form": "N per multiple unit",
"description": "A count over an unstated number of
days or months.",
"examples": [
  "1 per multiple day",
  "1 per multiple month"
]
},
{
  "form": "cluster per unit, N per cluster",
  "description": "How often clusters happen, and how
many seizures are in each cluster. The cluster count or
the time between clusters may be a range.",
  "examples": [
    "1 cluster per day, 5 per cluster",
    "1 cluster per week, 4 per cluster",
    "1 cluster per 4 month, 5 per cluster",
    "6 cluster per month, 4 per cluster",
    "1 cluster per 4 to 5 day, 2 per cluster"
  ]
},
{
  "form": "cluster per unit, range per cluster",
  "description": "How often clusters happen, and a
range of seizures in each cluster.",
  "examples": [
    "1 cluster per day, 2 to 4 per cluster",
    "1 cluster per week, 3 to 6 per cluster",
    "1 cluster per month, 4 to 6 per cluster"
  ]
},
{
  "form": "cluster per unit, multiple per cluster",
  "description": "How often clusters happen, and more
than one seizure in each cluster with no number given
.",
  "examples": [
    "1 cluster per day, multiple per cluster",
    "1 cluster per week, multiple per cluster",
    "1 cluster per month, multiple per cluster"
  ]
},
{
  "form": "unknown cluster count",
  "description": "Clusters are described, but how
often they happen is not known.",
  "examples": [
    "unknown, 5 per cluster",
    "unknown, 2 to 4 per cluster",
    "unknown, multiple per cluster"
  ]
},
{
  "form": "seizure free for a duration",
  "description": "No seizures for a stated length of
time.",
  "examples": [
    "seizure free for 6 month",
    "seizure free for 2 year",
    "seizure free for 3 months",
    "seizure free for 2 years",
    "seizure free for 2 to 3 months"
  ]
},
{
  "form": "seizure free for a vague duration",
  "description": "No seizures for an unstated length
of time.",

```

```

    "examples": [
      "seizure free for multiple month",
      "seizure free for multiple year"
    ],
  },
  {
    "form": "unknown",
    "description": "Seizures are discussed, but there is no usable frequency.",
    "examples": [
      "unknown"
    ]
  },
  {
    "form": "no seizure frequency reference",
    "description": "There is no usable frequency evidence.",
    "examples": [
      "no seizure frequency reference"
    ]
  }
],
"event_schema": {
  "event_id": "stable string such as e1",
  "kind": [
    "frequency_rate",
    "cluster_frequency",
    "seizure_free",
    "last_event_only",
    "unknown_frequency",
    "no_reference"
  ],
  "raw_value": "source-near expression or null",
  "applies_to": "seizure type or clinical target, or null",
  "time_window": "source-near current/recent/historical window, or null",
  "temporality": [
    "current",
    "recent",
    "historical",
    "future",
    "unclear"
  ],
  "assertion_status": [
    "asserted",
    "negated",
    "historical",
    "hypothetical",
    "unknown"
  ],
  "evidence": "exact note substring",
  "notes": "optional short note or null"
},
"selection_schema": {
  "selected_event_ids": "list of selected event_id strings",
  "final_kind": [
    "frequency",
    "seizure_free",
    "unknown",
    "no_reference",
    "unresolved_multiple"
  ],
  "final_label": "normalized label, or null if not directly countable",
  "evidence": "exact note substring supporting the final selection",
  "confidence": [
    "low",
    "medium",
    "high"
  ],
  "rationale": "brief clinical reason for selecting these events"
}
}

```

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